Session5: Penalized Linear Models

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Outline

1. Why penalization?
2. Penalized MRM
3. Penalized estimators

Error of train vs test

Under a general setting, assume we are interested in regressing a response variable (output of a process/machine) on a fixed covariate (input variables, some variables) through $Y = f(X) + \epsilon$, where $E(\epsilon) = 0$. Under the meaning of mean regression, $f(x) = E(Y|X = x)$. Note here, we do not consider non-parametric (or function) estimation and we will only focus on parametric regression, as for the case of multiple regression model.
Thus, the residual sum of squares (RSS) is given by

$$\mathrm{RSS}(f) = \sum_{i=1}^{n} (y_i - f(x_i))^2$$

for the set of sample points $\{(x_i, y_i)\}_{i=1}^{n}$.
I use Hastie et al. (2008)[1] as the reference.

[1] Hastie, T., Tibshirani, R. and Friedman, J. (2008) The Elements of Statistical Learning, 2nd Ed., Springer Series in Statistics

Error of train vs test cont.

Remember our ultimate goal of regression analysis is to build a model that predicts well.

First we need to take a measure for accounting "well". We start with the mean squared error (MSE), the expected value of the squared error loss function.

If we have enough data, we can divide the dataset into three parts: a training set, a validation set, and a test set. The training set is used to fit(build) the models; the validation set is used to estimate prediction error for model selection; the test set is used for assessment(check) of the generalization error of the final chosen model.

Error of train vs test cont.

Note

$$
\begin{aligned}
\mathrm{MSE}(\text{Estimate}) &= \mathrm{E}_{\mathrm{Train}}(L(f, \hat{Y})) \\
&= \mathrm{Var}(\hat{Y}) + \mathrm{Bias}(\hat{Y})^2 && (1.1) \\
\text{aka MPE} = \mathrm{MSE}(\text{Model}) &= \mathrm{E}_{\mathrm{Test}}(L(\boldsymbol{y}, \hat{Y})) \\
&= \mathrm{Var}(Y) + \mathrm{Var}(\hat{Y}) + \mathrm{Bias}(\hat{Y})^2 && (1.2) \\
&= \text{e1} + \text{e2} + \text{e3}
\end{aligned}
$$

where in (1.1), $\mathrm{Bias}(\hat{Y}) = \mathrm{E}(\hat{Y}(x_o) - f(x_o))$ over the train set, however, in (1.2) $\mathrm{Bias}(\hat{Y}) = \mathrm{E}(\hat{Y} - Y)$ over the test set.

Be careful with the training set—It can give you misleading evidence of model performance, that's why we use the test set to assess the model performance.

In (1.2):

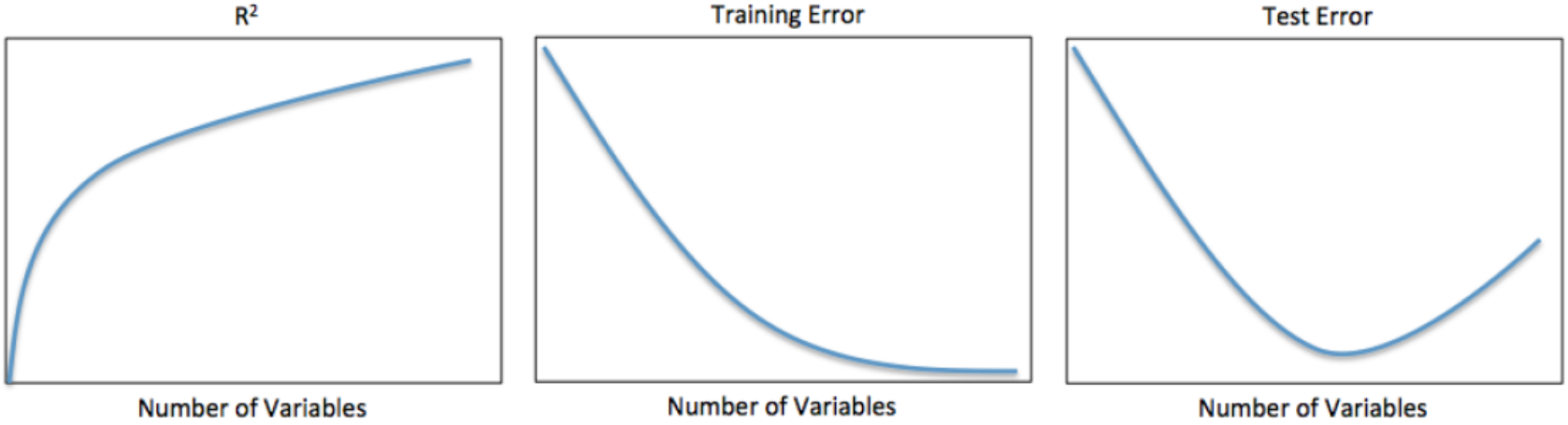
- e1 is called the irreducible error in Y which comes from the underlying, true model f. Even if our statistical model $\hat{f}$ exactly represented the real underlying model, it still contains this error.
- e2 is the variance of our estimated model $\hat{f}$. This is a random source of error in our model.
- e3 is the bias of our estimated model $\hat{f}$. This is a systematic source of error in our model.

Error of train vs test cont.

Once again, we use the training set to estimate the parameters. and If I use $\sum_{i=1}^{n_{\text{Train}}}(y_i - \hat{y}_i)^2$ for model assessment we may get misleading evidence. As the coefficient determination R^2 grows, the training error gets smaller (direct relation).

One way of increasing the R^2, is to add more variables. BUT, it will increase the model variance $\Longrightarrow$ overfitting.

Figure: Shojaie, A. (2017) High-dimensional statistical learning: Introduction, International Society for Business and Industrial Statistics, Bu Ali Sina University, Hamedan, Iran



Constrained estimation

Minimizing $\mathrm{RSS}(f)$ over the set of all functions f, leads to infinitely many solutions: any function $\hat{f}$ passing through the training points (x_i, y_i) is a solution. Any particular solution chosen might be a poor predictor at test points different from the training points.

In order to obtain useful results for finite n, we must restrict the eligible solutions to a smaller set of functions. Thus, we need to do a constrained estimation.

In short, we solve the following optimization problem, known as the penalized RSS

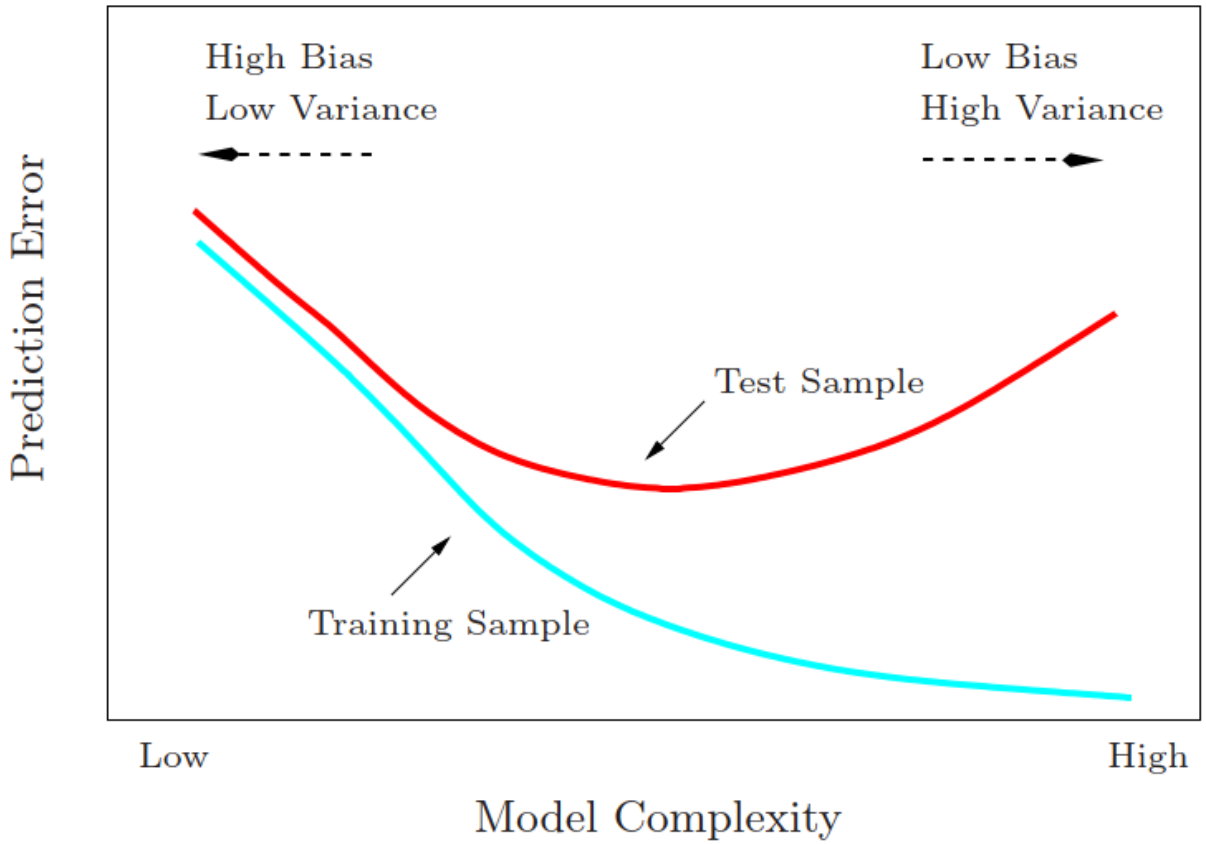
$$\min \mathrm{RSS}(f) + \lambda J(f). \tag{1.3}$$

Here, we term λ as the smoothing or complexity parameter, and $J(\cdot)$ is the penalty term.

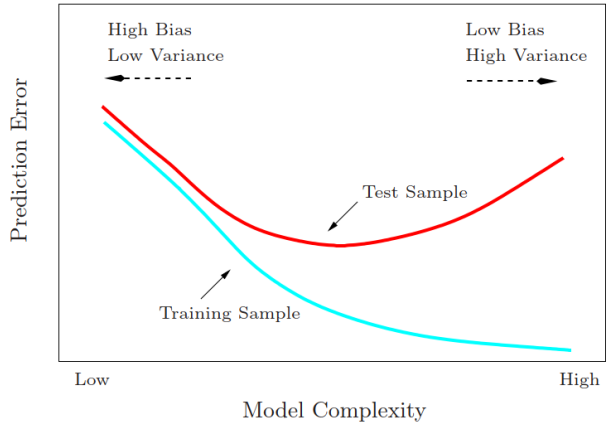
Bias-Variance tradeoff

The complexity parameter tunes the level of difficulty in estimation. (I will explain later)

Figure: Test and training error as a function of model complexity



Bias-Variance tradeoff cont.



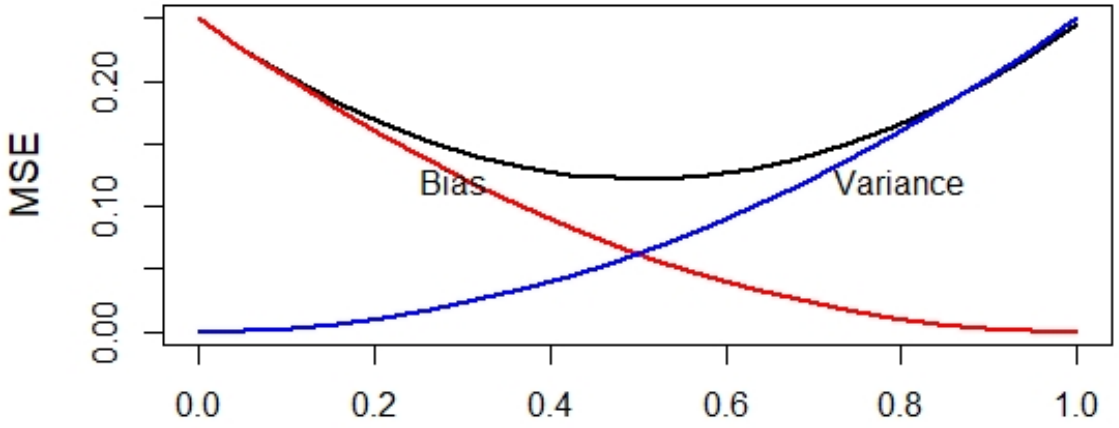
Fitting a model with many components (as we go right on the horizontal) leads to a model with low bias. This is because the model will be more complex and will therefore better fit the data. Note that, in this situation, we have more parameters to estimate, and this means that the variance of each parameter will be very large.

Fitting a simple model with few components (as we go left) gives a low variance model, because many data points will be available per parameter to estimate. However, we have a very parsimonious model and this will lead to greater bias.

Bias-Variance tradeoff cont.

Also, note that in the context of estimation, It is possible to find a model with lower MSE than an unbiased model! This is also called the "bias-variance tradeoff".

Figure: Bias-Variance tradeoff



The horizontal axis is the shrinkage factor???—We will come back.

Consider the multiple regression model

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \tag{2.1}$$

where $\boldsymbol{y}$ is an $n \times 1$ vector of responses, $\boldsymbol{X} = (\boldsymbol{x}_1, \cdots, \boldsymbol{x}_n)^\top$ is an $n \times p$ fixed-design matrix including the covariates $\boldsymbol{x}_i \in \mathbb{R}^p$, $i = 1, \cdots, n$, $\boldsymbol{\beta} = (\beta_1, \cdots, \beta_p) \in \mathbb{R}^p$ is the regression coefficients' vector, and $\boldsymbol{\epsilon}$ is an $n \times 1$ vector of measurement errors (we eliminated the intercept from our model) with $\mathrm{E}(\boldsymbol{\epsilon}) = \mathbf{0}$.

If the columns of the design matrix $\boldsymbol{X}$ are linearly independent, then the estimator of $\boldsymbol{\beta}$ is the unique solution $\boldsymbol{b} = \hat{\boldsymbol{\beta}}$ of the LS problem (see Session 1)

$$\min \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}\|_2^2 \quad \text{subject to} \quad \boldsymbol{b} \in \mathbb{R}^p, \tag{2.2}$$

where for $\boldsymbol{a} = (a_1, \cdots, a_n)^\top$, $\|\boldsymbol{a}\|_p = \left(\sum_{i=1}^n a_i^p\right)^{\frac{1}{p}}$.

Recall from Session 1, if we have a high-dimensional case, or rank deficient design matrix, or a multicollinear situation, then, we need to take an action for the OLS estimate.

Let's focus on not-independent columns case: I elaborate more.

Under situations in which the matrix $\boldsymbol{X}^\top \boldsymbol{X}$ is ill-conditioned due to near/exact linear relationship among the regressors of $\boldsymbol{X}$, we always find a linear combination of the columns in $\boldsymbol{X}$ which is exactly equal to one other (not full rank), and one may have $\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{X}(\boldsymbol{\beta} + \boldsymbol{\zeta})$ for every $\boldsymbol{\zeta}$ in the null space of $\boldsymbol{X}$. It is then impossible to separate the contribution of the individual covariates. The uncertainty with respect to the covariate responsible for the variation explained in $\boldsymbol{y}$ is often reflected in the fit of the linear regression model to data by a large error of the estimates of the regression parameters corresponding to the collinear covariates and, consequently, usually result large values of the estimates.

Consequently, without applying constraint, it is impossible to infer or estimate $\boldsymbol{\beta}$ from data.

It is not hard to decide which constraint. We want to estimate $\boldsymbol{\beta}$, thus make the constraint on $\boldsymbol{\beta}$. Intuitively, you may think that the magnitude of $\boldsymbol{\beta}$ causes the problem (since $\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{X}(\boldsymbol{\beta} + \boldsymbol{\zeta})$). Thus, make a constraint based on the magnitude of $\boldsymbol{\beta}$ and since it is a vector, use a norm. Therefore, the penalized regression problem has form

Constrained form

$$\min \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}\|_2^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\| \leq c, \quad c > 0. \tag{2.3}$$

Extremely important: n **is the number of observations in the training set**, if we have enough sample points to divide into training and test sets.

Penalized least squares estimator

Equivalently, using the Lagrangian multipliers, the solution to the constrained LS problem is given by (equivalently)

Lagrange multiplier form

$$\hat{\boldsymbol{\beta}}_{\lambda}^{\mathrm{Pen}} = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\| \right\} \tag{2.4}$$

You like this one $\downarrow$ more!!!

$$\hat{\boldsymbol{\beta}}_{\lambda}^{\mathrm{Pen}} = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \sum_{i=1}^{p} p_\lambda(|\beta_j|) \right\} \tag{2.5}$$

which we use $p_\lambda(|\beta_j|) = \lambda p(|\beta_j|)$ (often).

Relation between c and λ

What is the relation/connection between the upper bound condition parameter c and the tuning (shrinkage) parameter λ?
Consider $p(x) = x$. Then, $\sum_{i=1}^{p} |\beta_i| \leq c$ shrinks the coefficients and as c gets smaller the amount of shrinkage increases. For the case $c \to 0$, $\beta_i \to 0$, $\forall\ i = 1, \cdots, p$. In contrast, As λ increases, then $\sum_{i=1}^{p} |\beta_i|$ decreases, i.e. the coefficients become shrunken towards zero. Thus, c and λ have reverse (contrary) performance.

Now the questions are: Which penalty function? How to select λ?

Subset selection

Recall the purpose again! Two reasons that we are not happy with the OLS (Hastie et al., 2008)

- The first is *prediction accuracy*: the least squares estimates often have low bias but large variance. Prediction accuracy can sometimes be improved by shrinking or setting some coefficients to zero. By doing so we sacrifice a little bit of bias to reduce the variance of the predicted values, and hence may improve the overall prediction accuracy.

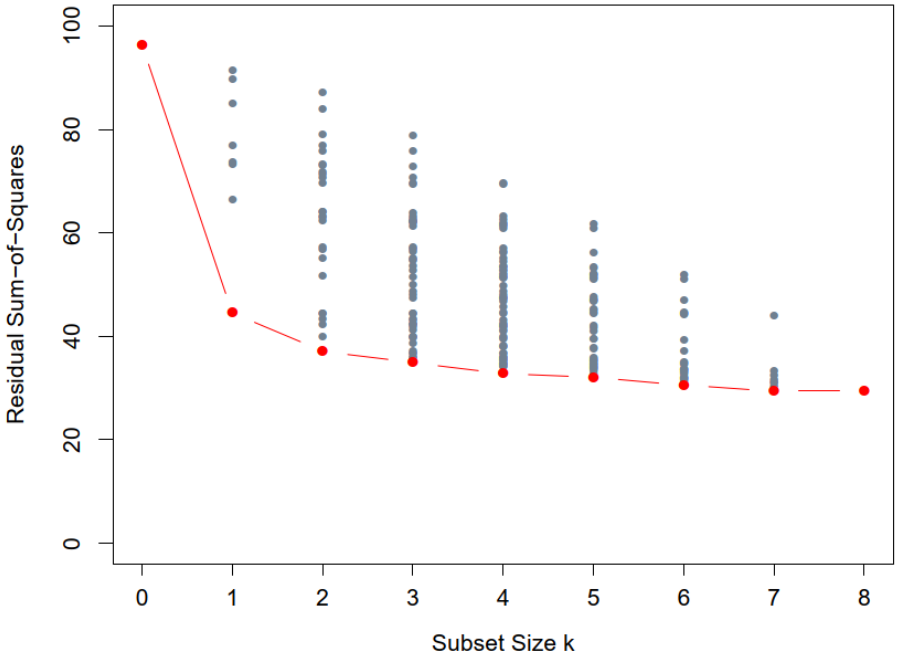
- The second reason is *interpretation*. With a large number of predictors, we often would like to determine a smaller subset that exhibit the strongest effects. In order to get the "big picture," we are willing to sacrifice some of the small details.

Thus, we do not want a complex model, in which many predictors contribute in estimation $\boldsymbol{y}$; parsimony is important.

Subset selection cont.

Best subset regression finds for each $k \in \{0, 1, 2, \cdots, p\}$ the subset of size k that gives smallest residual sum of squares

$$\mathrm{RSS}(\boldsymbol{\beta}) = \sum_{i=1}^{n}(y_i - \sum_{j=1}^{p} x_{ij}\beta_j)^2 \tag{2.6}$$



This figure shows all the subset models for a real data example. The lower boundary represents the models that are eligible for selection by the best-subsets approach. Note that the best subset of size 2, for example, need not include the variable that was in the best subset of size 1. The best-subset curve (red lower boundary) is necessarily decreasing, so cannot be used to select the subset size k. The question of how to choose k involves the bias-variance tradeoff. Typically we choose the smallest model that minimizes an estimate of the expected prediction error.

Subset selection cont.

Thus, the subset selection can be viewed as a possible remedy for the OLS unhappiness. Can we formulate subset selection as a penalized problem? Simply, the objective function is

$$\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda p. \tag{2.7}$$

Remember we have $2^p - 1$ possible combination of variables. (why?)
Note, the other known model selection methods are also penalized procedures, e.g., AIC, BIC, Mallows' C_p.

Good penalty

Before we consider some well-known and useful penalty functions for the question "Which penalty function?", we define a good penalty function. According to Fan and Li (2001)[2], any good penalty function must have the following properties:

- Unbiasedness
- Sparsity
- Continuity

Thus, our target is to focus on penalty functions with good properties. However, it is not always achievable and sometimes we use a penalty that fulfills only one property (e.g. continuity in the L_2-norm case as in the ridge penalty).

[2] Fan, J. and Li, R. (2001) Variable selection via nonconcave penalized likelihood and its oracle properties, J. Amer. Statist. Assoc., 96(456), 1348-1360

Good penalty cont.

Unbiasedness The resulting estimator is nearly unbiased when the true unknown parameter is large to avoid unnecessary modeling bias. Thus, we need $p_\lambda(\cdot)$ that results in an estimator that that is nearly unbiased for large components of $\boldsymbol{\beta}$.

Sparsity The resulting estimator is a thresholding rule, which automatically sets small estimated coefficients to zero to reduce model complexity. Thus, we need $p_\lambda(\cdot)$ that results in an estimator that sets some components of $\boldsymbol{\beta}$ to 0.

Continuity The resulting estimator is continuous in $(\boldsymbol{X}^\top \boldsymbol{y})$ data in the training set to avoid instability in model prediction.

Good penalty cont.

In order to digest the good properties and a way to check, we consider the orthonormal design case in which $\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{X}\boldsymbol{X}^\top = \boldsymbol{I}_p$. Then,

$$\begin{aligned}
\|\boldsymbol{y} - \boldsymbol{X}\beta\|_2^2 &= \boldsymbol{y}^\top \boldsymbol{y} - 2\beta^\top \boldsymbol{X}^\top \boldsymbol{y} + \beta^\top \boldsymbol{X}^\top \boldsymbol{X}\beta \\
&= \boldsymbol{y}^\top \boldsymbol{y} - 2\beta^\top \boldsymbol{X}^\top \boldsymbol{y} + \beta^\top \boldsymbol{X}^\top \boldsymbol{X}\beta + 2\boldsymbol{y}^\top \boldsymbol{X}\boldsymbol{X}^\top \boldsymbol{y} - 2\boldsymbol{y}^\top \boldsymbol{X}\boldsymbol{X}^\top \boldsymbol{y} \\
&= \boldsymbol{y}^\top \boldsymbol{y} - 2\boldsymbol{y}^\top \boldsymbol{X}\boldsymbol{X}^\top \boldsymbol{y} + \boldsymbol{y}^\top \boldsymbol{X}\boldsymbol{X}^\top \boldsymbol{X}\boldsymbol{X}^\top \boldsymbol{y} + \boldsymbol{y}^\top \boldsymbol{X}\boldsymbol{X}^\top \boldsymbol{y} \\
&\quad -2\beta^\top \boldsymbol{X}^\top \boldsymbol{y} + \beta^\top \boldsymbol{X}^\top \boldsymbol{X}\beta \\
&= \boldsymbol{y}^\top (\boldsymbol{I} - \boldsymbol{X}\boldsymbol{X}^\top)^\top (\boldsymbol{I} - \boldsymbol{X}\boldsymbol{X}^\top)\boldsymbol{y} + (\boldsymbol{X}^\top \boldsymbol{y} - \beta)^\top (\boldsymbol{X}^\top \boldsymbol{y} - \beta) \\
&= \|\boldsymbol{y} - \hat{\boldsymbol{y}}\|_2^2 + \|\hat{\beta} - \beta\|_2^2. \qquad (2.8)
\end{aligned}$$

Note here, $\hat{\beta} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{y} = \boldsymbol{X}^\top \boldsymbol{y}$ and $\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\beta} = \boldsymbol{X}\boldsymbol{X}^\top \boldsymbol{y}$.
Let's take $\boldsymbol{z} = \boldsymbol{X}^\top \boldsymbol{y} = (z_1, \cdots, z_p)$.

Good penalty cont.

Thus, minimizing $\frac{1}{2}\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta}\|_2^2+\sum_{i=1}^p p_\lambda(|\beta_j|)$ is equivalent to minimizing

$$\frac{1}{2}(z_j-\beta_j)^2+p_\lambda(|\beta_j|) \tag{2.9}$$

elementwise for each β_j, $j=1,\cdots,p$.

First note sometimes to avoid unnecessary coefficient factor, we use $\frac{1}{2}\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta}\|_2^2$, or $\frac{1}{n}\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta}\|_2^2$, or $\frac{1}{2n}\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta}\|_2^2$, instead. Further, in sequel, I ignore the subscript j and use z and β instead of z_j and β_j, for the ease of understanding

To minimize the new objective function, we get derivatives. Thus, the fist order derivative of (2.9) is (ignoring j)

$$\begin{aligned} & -z+\beta+p'_\lambda(|\beta|)+\mathrm{sgn}(\beta) \\ = \quad & \mathrm{sgn}(\beta)\left(|\beta|+p'_\lambda(|\beta|)\right)-z \\ = \quad & q(\beta) \quad (\text{say}) \end{aligned} \tag{2.10}$$

Then we seek for β such that $q(\beta)=0$.

Good penalty cont.

Unbiasedness Since the OLS estimator (z, here) is unbiased, we require for large values β, our estimator becomes equal to z. From (2.10), it is easy to see that when $p'_\lambda(|\beta|) = 0$, for large $|\beta|$, then the resulting estimator is z ($\beta = z$). Thus the condition for unbiased0ness is

$$p'_\lambda(|\beta|) \to 0 \quad \text{as} \quad |\beta| \to \infty. \tag{2.11}$$

Good penalty cont.

Sparsity In this case $q(0) = 0$. It means we have $\beta = 0$ that minimizes (2.9). This satisfies if $q(\beta)$ changes sign as β does at the origin.

$$\begin{aligned} q(\beta) > 0 \quad &\Longrightarrow \quad z < \operatorname{sgn}(\beta)\left(|\beta| + p'_\lambda(|\beta|)\right) \\ &\overset{\beta>0}{\Longrightarrow} \quad z < \inf_\beta \left(|\beta| + p'_\lambda(|\beta|)\right) \\ q(\beta) < 0 \quad &\Longrightarrow \quad z > \operatorname{sgn}(\beta)\left(|\beta| + p'_\lambda(|\beta|)\right) \\ &\overset{\beta<0}{\Longrightarrow} \quad -z < \inf_\beta \left(|\beta| + p'_\lambda(|\beta|)\right) \end{aligned}$$

Thus we need $|z| < \inf_\beta \left(|\beta| + p'_\lambda(|\beta|)\right)$. This condition is satisfied if

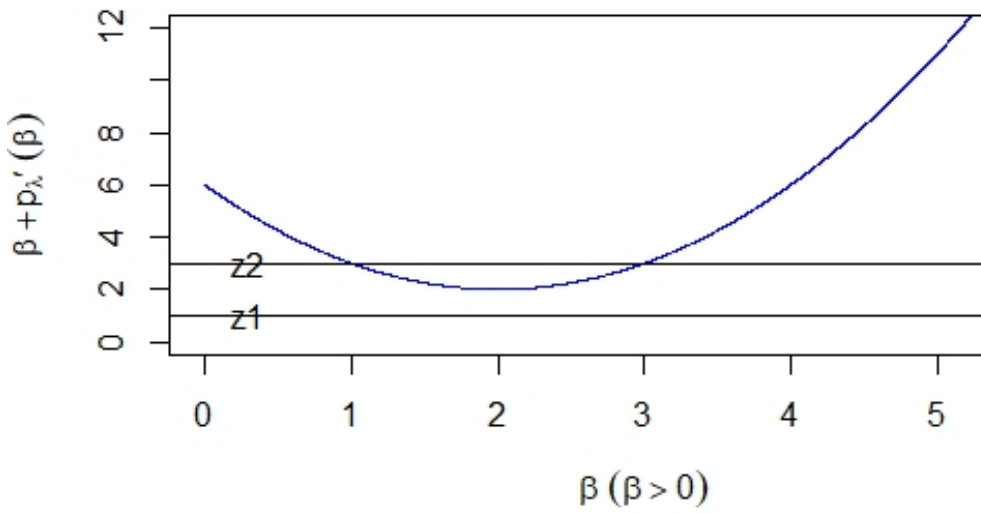
$$\inf_\beta \left(|\beta| + p'_\lambda(|\beta|)\right) > 0 \tag{2.12}$$

since $|z| > 0$ always.

Good penalty cont.

Continuity Continuity means if we change the input data, we must not have jump. For $|z| < \inf_\beta (|\beta| + p'_\lambda(|\beta|))$, β is shrunken to zero. For $|z| > \inf_\beta (|\beta| + p'_\lambda(|\beta|))$[3] two crossings may exist as shown in the this figure; the larger one is a penalized least squares estimator. This implies $\beta = 0$ must be minimizer of $(|\beta| + p'_\lambda(|\beta|))$. Thus

$$\arg\inf_\beta \left(|\beta| + p'_\lambda(|\beta|)\right) = 0 \quad (2.13)$$



[3] Note here, the training data z is changed

Ridge regression

As you are familiar (from Session 1), the ridge regression comes first. Using the L_2-norm, take $p(|\beta_j|) = \beta_j^2$. Thus, we solve the following constraint problem

$$\min \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|_2^2 \leq c, \quad c > 0. \tag{3.1}$$

where $\|\boldsymbol{\beta}\|_2^2 = \boldsymbol{\beta}^\top \boldsymbol{\beta} = \sum_{i=1}^p \beta_j^2$. Thus,

$$\hat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda) = \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left\{ PS_\lambda^{\text{ridge}}(\boldsymbol{\beta}) \right\}, \tag{3.2}$$

where the penalized least squares criterion is

$$PS_\lambda^{\text{ridge}}(\boldsymbol{\beta}) = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2. \tag{3.3}$$

Ridge regression cont.

From (for both low and high dimensional cases, you know from Session 1)

$$\frac{\partial PS_\lambda^{\text{ridge}}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \frac{\partial}{\partial \boldsymbol{\beta}} \boldsymbol{y}^\top \boldsymbol{y} - 2\boldsymbol{\beta}^\top \boldsymbol{X}^\top \boldsymbol{y} + \boldsymbol{\beta}^\top \boldsymbol{X}^\top \boldsymbol{X}\boldsymbol{\beta} + \lambda \boldsymbol{\beta}^\top \boldsymbol{\beta} = -2\boldsymbol{X}^\top \boldsymbol{\beta} + 2\boldsymbol{X}^\top \boldsymbol{X}\boldsymbol{\beta} + 2\lambda\boldsymbol{\beta} = \boldsymbol{0} \tag{3.4}$$

the ridge estimator has form

$$\hat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda) = (\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I}_p)^{-1} \boldsymbol{X}^\top \boldsymbol{y}. \tag{3.5}$$

If $\boldsymbol{X}^\top \boldsymbol{X}$ is invertible, using the OLS estimator $\hat{\boldsymbol{\beta}}$, we have

$$\begin{aligned} \hat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda) &= (\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I}_p)^{-1} (\boldsymbol{X}^\top \boldsymbol{X})(\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y} \\ &= \boldsymbol{C}(\lambda)^{-1} \hat{\boldsymbol{\beta}}, \quad \boldsymbol{C}(\lambda) = (\boldsymbol{I}_p + \lambda (\boldsymbol{X}^\top \boldsymbol{X})^{-1}). \end{aligned} \tag{3.6}$$

Remember the discussion in Session 1 for the orthonormal case.

Ridge regression cont.

Augmented data construction: what is the benefit? In any context the OLS is valid, you can get the ridge using augmentation.
Consider the following augmented data (Saleh et al., 2019)

$$\boldsymbol{X}^* = \begin{bmatrix} \boldsymbol{X} \\ \sqrt{k}\boldsymbol{I}_p \end{bmatrix}, \quad \boldsymbol{Y}^* = \begin{bmatrix} \boldsymbol{Y} \\ \boldsymbol{0} \end{bmatrix}. \tag{3.7}$$

Assume the following multiple regression model,

$$\boldsymbol{Y}^* = \boldsymbol{X}^*\boldsymbol{\beta} + \boldsymbol{\epsilon}^*, \tag{3.8}$$

where $\boldsymbol{\epsilon}^*$ is an $(n+p)$-vector of i.i.d. random variables. Then, the OLS estimator of $\boldsymbol{\beta}$ is obtained as

$$\begin{aligned} \boldsymbol{\beta}_n^* &= \min_{\boldsymbol{\beta}} \left\{ (\boldsymbol{Y}^* - \boldsymbol{X}^*\boldsymbol{\beta})^\top (\boldsymbol{Y}^* - \boldsymbol{X}^*\boldsymbol{\beta}) \right\} \\ &= (\boldsymbol{X}^{*\top}\boldsymbol{X}^*)^{-1}\boldsymbol{X}^{*\top}\boldsymbol{Y}^* \\ &= (\boldsymbol{X}^\top\boldsymbol{X} + k\boldsymbol{I}_p)^{-1}\boldsymbol{X}^\top\boldsymbol{Y} \\ &= \hat{\boldsymbol{\beta}}^{\text{ridge}}(k). \end{aligned}$$

Ridge regression cont.

There always exists a $\lambda > 0$ such that the risk of $\hat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda)$ is lesser than that of the OLS estimator. How should we find(estimate) λ then?

(Saleh et al., 2019) Define the generalized cross validation (GCV) as the weighted average of predicted square errors

$$\mathrm{Var}(\lambda) = \frac{1}{n}\sum_{i=1}^{n}\left(\boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda) - y_i\right)^2 w_i(\lambda), \tag{3.9}$$

where

$$w_i(\lambda) = \frac{(1 - a_{ii}(\lambda))}{1 - \frac{1}{n}\operatorname{tr}\boldsymbol{A}(\lambda)}, \quad \boldsymbol{A}(\lambda) = \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X} + \lambda\boldsymbol{I}_p)\boldsymbol{X}^\top, \tag{3.10}$$

where $a_{ii}(\lambda)$ is the ith diagonal element of $\boldsymbol{A}(\lambda)$.

Ridge regression cont.

A computationally efficient version of the GCV function is

$$\mathrm{GCV}(\lambda) = \frac{1}{n} \frac{\|(\boldsymbol{I}_n - \boldsymbol{A}(\lambda))\boldsymbol{y}\|_2^2}{\left[\frac{1}{n}\operatorname{tr}(\boldsymbol{I}_n - \boldsymbol{A}(\lambda))\right]^2} \tag{3.11}$$

The GCV estimator of λ is then given by

$$\lambda_{\mathrm{opt}} = \arg\min_{\lambda>0} \mathrm{GCV}(\lambda).$$

Ridge regression cont.

Ridge trace is a two dimensional plot of $\hat{\beta}_j^{\text{ridge}}(\lambda)$ and the RSS given by

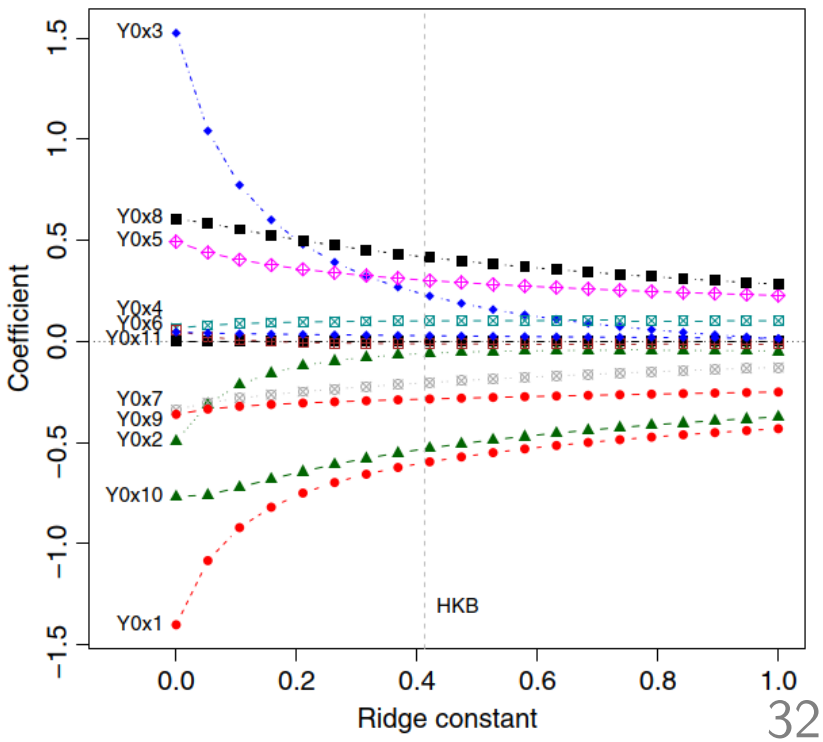
$$\text{RSS}(k) = \left[\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda)\right]^\top \left[\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda)\right],$$

for a number of values of $\lambda \in [0, 1]$. The trace serves to depict the complex relationships that exist between non orthonormal prediction vectors and the effect of these interrelationships on the estimation of $\boldsymbol{\beta}$. Also, it is the way to estimate λ_{opt}.

Ridge regression cont.

Now, we discuss the gasoline mileage data from Montgomery et al. (2012). The data contains the following variables: $y =$ Miles/gallon, $x_1 =$ Displacement (cubic in.), $x_2 =$ Horsepower (ft-lb), $x_3 =$ Torqne (ft-lb), $x_4 =$ Compression ratio, $x_5 =$ Rear axle ratio, $x_6 =$ Carburetor (barrels), $x_7 =$ Number of transmission speeds, $x_8 =$ Overall length, $x_9 =$ E=Width (in), $x_{10} =$ Weight (lb), $x_{11} =$ Type of transmission. We may find that the eigenvalues of the matrix $\boldsymbol{X}^\top \boldsymbol{X}$ are given by $\lambda_1 = 223.33$, $\lambda_2 = 40.70$, $\lambda_3 = 22.43$, $\lambda_4 = 16.73$, $\lambda_5 = 6.13$, $\lambda_6 = 4.11$, $\lambda_7 = 2.76$, $\lambda_8 = 1.45$, $\lambda_9 = 0.96$, $\lambda_{10} = 0.24$, and $\lambda_{11} = 0.11$.

The condition number is $\kappa = 223.338/0.110 = 2030.345$, which indicates serious multicollinearity. From the figure, we can see that reasonable coefficient stability is achieved in the range $4.0 < \lambda_{\text{opt}} < 4.2$. Note, the HKB is the (3.10) in Session 1.



Ridge regression cont.

Ridge is important!!!

Package	Description
lmridge	A comprehensive R package for ridge regression
lrmest	Computes estimators such as OLS, ordinary RR, Liu
ltsbase	Computes ridge and Liu estimates based on the least trimmed squares method
penalized	Computes LASSO, ridge and fused LASSO
glmnet	Computes LASSO, ridge, and elastic-net (we used in Session 1)
ridge	Computes ridge for linear and logistic models for genome-wide SNP data
RXshrink	Computes generalized ridge estimator using the ML method
rrBLUP	Computes ridge estimator for genomic selection
RidgeFusion	Computes ridge estimator for fusion statistical learning, QDA, and cluster analysis

Let's work on "lmridge: A Comprehensive R Package for Ridge Regression"
We study the pdf file:
D://Joint Researches/Prof. Bekker/LMO720/Sessions

LASSO

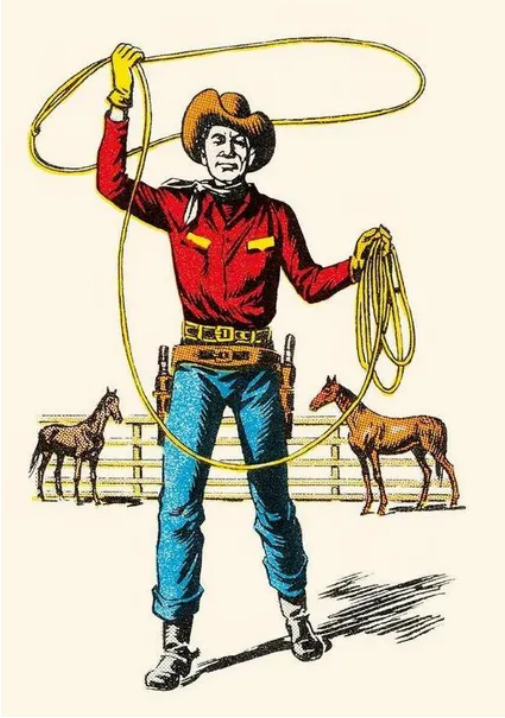
Major drawbacks of the ridge:
(for the orthonormal design, since $p_\lambda(|\beta|) = \lambda\beta^2$)

$$p'_\lambda(|\beta|) = 2\lambda\beta \to \infty \quad \text{as} \quad \beta \to \infty \Longrightarrow \text{biased}$$
$$\inf_\beta (|\beta| + 2\lambda\beta) = 0 \Longrightarrow \text{not sparse}$$
$$\arg\inf_\beta (|\beta| + 2\lambda\beta) = 0 \Longrightarrow \text{continues}$$

no feature selection; estimate all the p features, thus no parsimony; in high-dimension, does not give an interpretable model.

Tibshirani (1996) proposed the least absolute shrinkage and selection operator (LASSO).

Tibshirani, R. (1996) Regression shrinkage and selection via the lasso, J. Royal Statist. Soc. Ser. B, 58(1), 267-288.



LASSO

Using the L_1-norm, take $p(|\beta_j|) = |\beta_j|$. Thus, we solve the following constraint problem

$$\min \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|_1 \leq c, \quad c > 0, \tag{3.12}$$

where $\|\boldsymbol{\beta}\|_1 = \sum_{j=1}^{p} |\beta_j|$. Thus,

$$\hat{\boldsymbol{\beta}}^{\text{LASSO}}(\lambda) = \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left\{ PS_\lambda^{\text{LASSO}}(\boldsymbol{\beta}) \right\}, \tag{3.13}$$

where the penalized least squares criterion is

$$PS_\lambda^{\text{LASSO}}(\boldsymbol{\beta}) = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1. \tag{3.14}$$

Note $\|\boldsymbol{\beta}\|_1 = \mathbf{1}_p^\top |\boldsymbol{\beta}|$, where $\mathbf{1}_p$ is a p-tuple of 1's and $|\boldsymbol{\beta}| = (|\beta_1|, \cdots, |\beta_p|)^\top$.

LASSO cont.

There is no closed form solution for (3.13). Now, consider the orthonormal design and take $PS_{\lambda}^{\mathrm{LASSO}}(\boldsymbol{\beta}) = \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1$. See Saleh et al. (2019) for non-orthonormal case!
Differentiating $PS_{\lambda}^{\mathrm{LASSO}}(\boldsymbol{\beta})$ and setting the resultant to zero, we obtain ($\boldsymbol{X}^\top\boldsymbol{X} = \boldsymbol{I}_p$)

$$-\boldsymbol{X}^\top\boldsymbol{y} + \boldsymbol{\beta} + \lambda\mathrm{sgn}(\boldsymbol{\beta}) = \mathbf{0} \Longrightarrow -\hat{\boldsymbol{\beta}} + \boldsymbol{\beta} + \lambda\mathrm{sgn}(\boldsymbol{\beta}) = \mathbf{0}, \qquad (3.15)$$

where $\mathrm{sgn}(\boldsymbol{\beta}) = (\mathrm{sgn}(\beta_1), \ldots, \mathrm{sgn}(\beta_p))^\top$. Now, the jth marginal component of (3.15) is given by $\hat{\beta}_j^{\mathrm{LASSO}}(\lambda)$,

$$-\hat{\beta}_j + \hat{\beta}_j^{\mathrm{LASSO}}(\lambda) + \lambda\mathrm{sgn}\left(\hat{\beta}_j^{\mathrm{LASSO}}(\lambda)\right) = 0;\ j = 1, 2, \ldots, p. \qquad (3.16)$$

LASSO cont.

Now, we have three cases:

(i) $\operatorname{sgn}\left(\hat{\beta}_j^{\mathrm{LASSO}}(\lambda)\right) = +1$, then (3.16) reduces to

$$-\hat{\beta}_j + \hat{\beta}_j^{\mathrm{LASSO}}(\lambda) + \lambda = 0, \quad j = 1, 2, \ldots, p. \tag{3.17}$$

Hence,

$$0 < \hat{\beta}_j^{\mathrm{LASSO}}(\lambda) = \left(\hat{\beta}_j - \lambda\right) = \left|\hat{\beta}_j\right| - \lambda, \tag{3.18}$$

with, clearly, $\hat{\beta}_j > 0$ and $\left|\hat{\beta}_j\right| > \lambda$.

LASSO cont.

(ii) $\text{sgn}\left(\hat{\beta}_j^{\text{LASSO}}(\lambda)\right) = -1$, then we have

$$-\hat{\beta}_j + \hat{\beta}_j^{\text{LASSO}}(\lambda) - \lambda = 0; \quad j = 1, 2, \ldots, p. \tag{3.19}$$

Hence,

$$\begin{aligned} 0 > \hat{\beta}_j^{\text{LASSO}}(\lambda) &= \hat{\beta}_j + \lambda = -\left|\hat{\beta}_j\right| + \lambda \\ \hat{\beta}_j^{\text{LASSO}}(\lambda) &= -\left(\left|\hat{\beta}_j\right| - \lambda\right) \end{aligned} \tag{3.20}$$

with, clearly, $\hat{\beta}_j < 0$ and $\left|\hat{\beta}_j\right| > \lambda$.

LASSO cont.

(iii) For $\hat{\beta}_j^{\mathrm{LASSO}}(\lambda) = 0$, we have $-\hat{\beta}_j + \lambda\gamma = 0$ for some $\gamma \in [-1, 1]$. Hence, we obtain $\hat{\beta}_j = \lambda\gamma$, which implies $\left|\hat{\beta}_j\right| \leq \lambda$.

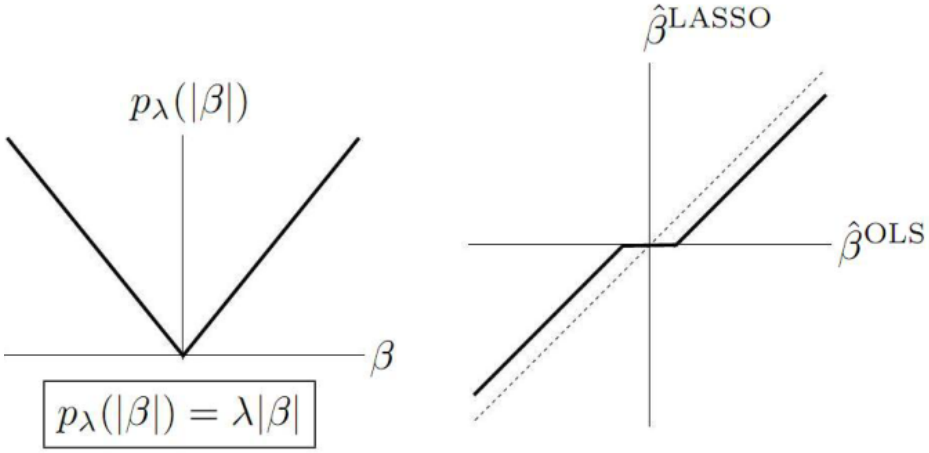
Combining (3.18), (3.20), and (iii), we obtain the LASSO of Tibshirani (1996) for the orthonormal case as

$$\hat{\beta}_j^{\mathrm{LASSO}}(\lambda) = \mathrm{sgn}(\hat{\beta}_j)\left(|\hat{\beta}_j| - \lambda\right)_+, \tag{3.21}$$

where $a_+ = \max(0, a)$ and $\hat{\beta}_j$ is the jth element of the OLS estimator $\hat{\boldsymbol{\beta}}$.

LASSO cont.

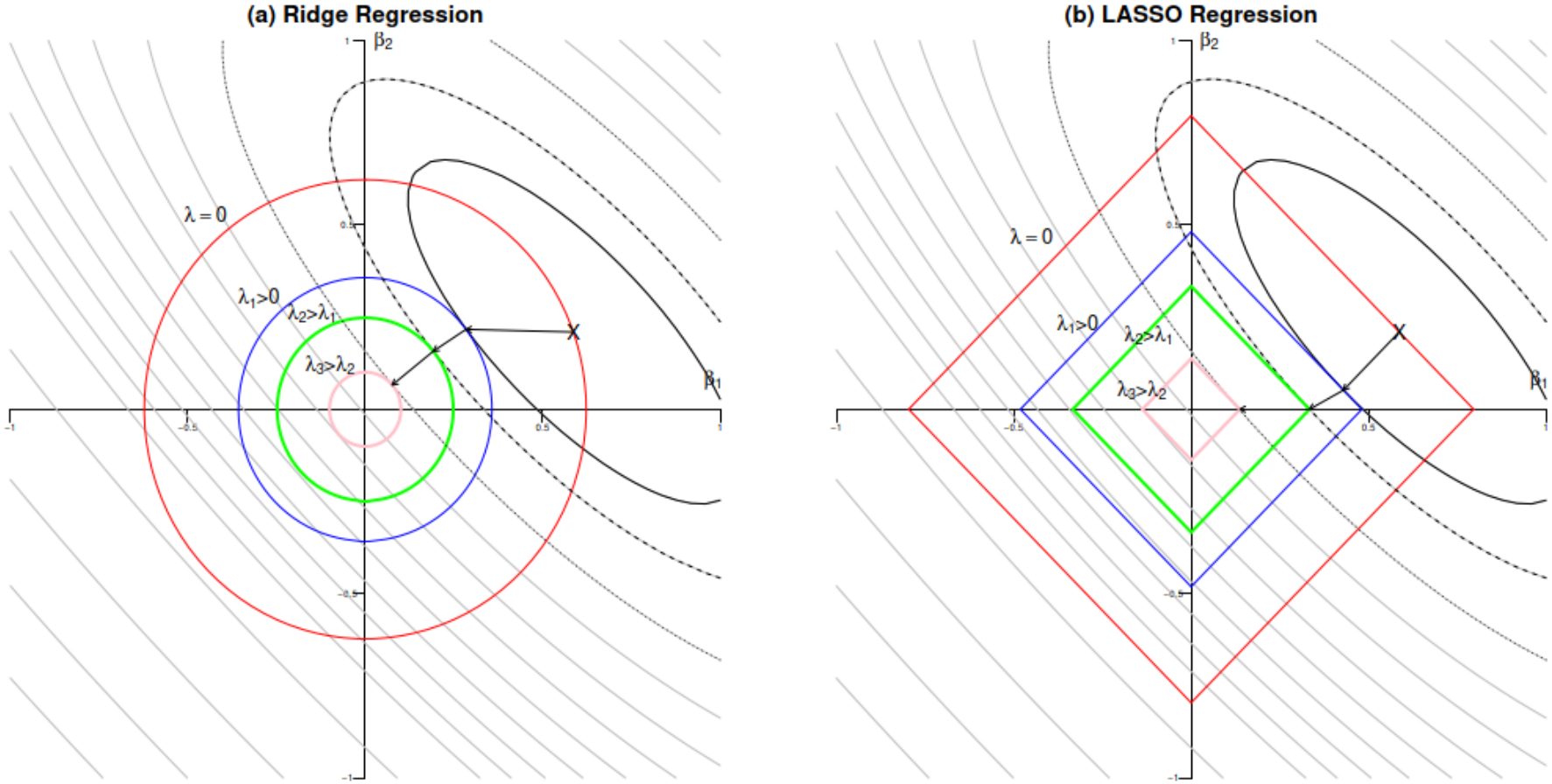
LASSO is known as the soft thresholding method. What is hard thresholding?



It is clear how the LASSO biases the estimates of large values of β.
It can be shown that LASSO selects the variables and gives sparse solution.
Let's elaborate more $\longrightarrow$

LASSO cont.

Bivariate geometrical comparison, i.e., the case $p = 2$ with $\boldsymbol{\beta} = (\beta_1, \beta_2)^\top$.



Lab12

See van Wieringen (2020)[4] pdf file. I strongly suggest to read more.
D://Joint Researches/Prof. Bekker/LMO720/Sessions
Simulation study adopted from the above reference.
non- and uniqueness of the lasso-estimator and predictor: We generate the data according to $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ where the $n = 5$ rows of $\boldsymbol{X}$ are sampled from $N_p[\mathbf{0}, (1-\rho)\boldsymbol{I}_p + \rho\mathbf{1}_p]$[5] with $p = 10$, $\rho = 0.99$, $\boldsymbol{\beta} = (\mathbf{1}_3^\top, \mathbf{0}_7^\top)^\top$ and $\boldsymbol{\epsilon} \sim N_5(\mathbf{0}_5, \frac{1}{10}\boldsymbol{I}_5)$. We choose $\lambda = \lambda_1 = 1$. We use the packages "penalized" and "glmnet". Both estimates, denoted $\hat{\boldsymbol{\beta}}_p(\lambda)$ and $\hat{\boldsymbol{\beta}}_g(\lambda)$.

[4] van Wieringen, W.N. (2020) Lecture notes on ridge regression, arXiv:1509.09169v6.
[5] Note we have an exchangeable covariance matrix.

Lab12 cont.

		β_1	β_2	β_3	β_4	β_5	β_6	β_7	β_8	β_9	β_{10}
`penalized`	$\hat{\boldsymbol{\beta}}_{\mathrm{p}}(\lambda_1)$	0.267	0.000	1.649	0.093	0.000	0.000	0.000	0.571	0.000	0.269
`glmnet`	$\hat{\boldsymbol{\beta}}_{\mathrm{g}}(\lambda_1)$	0.269	0.000	1.776	0.282	0.195	0.000	0.000	0.325	0.000	0.000

This output is troublesome when it comes to communication of the optimal model. From a different perspective the realized loss $PS_{\lambda_1}^{\mathrm{LASSO}}(\boldsymbol{\beta})$ for each estimate is approximately equal to 2.99, with the difference possibly due to convergence criteria of the algorithms. On another note, their corresponding predictors, $\boldsymbol{X}\hat{\boldsymbol{\beta}}_p(\lambda_1)$ and $\boldsymbol{X}\hat{\boldsymbol{\beta}}_g(\lambda_1)$, correlate almost perfectly: $\mathrm{corr}[\boldsymbol{X}\hat{\boldsymbol{\beta}}_p(\lambda_1), \boldsymbol{X}\hat{\boldsymbol{\beta}}_g(\lambda_1)] = 0.999$. These results thus corroborate the non-uniqueness of the estimator and the uniqueness of the predictor.

Lab13

Adopted from Efron et al. (2004)[6]

TABLE 1

Diabetes study: 442 diabetes patients were measured on 10 baseline variables; a prediction model was desired for the response variable, a measure of disease progression one year after baseline

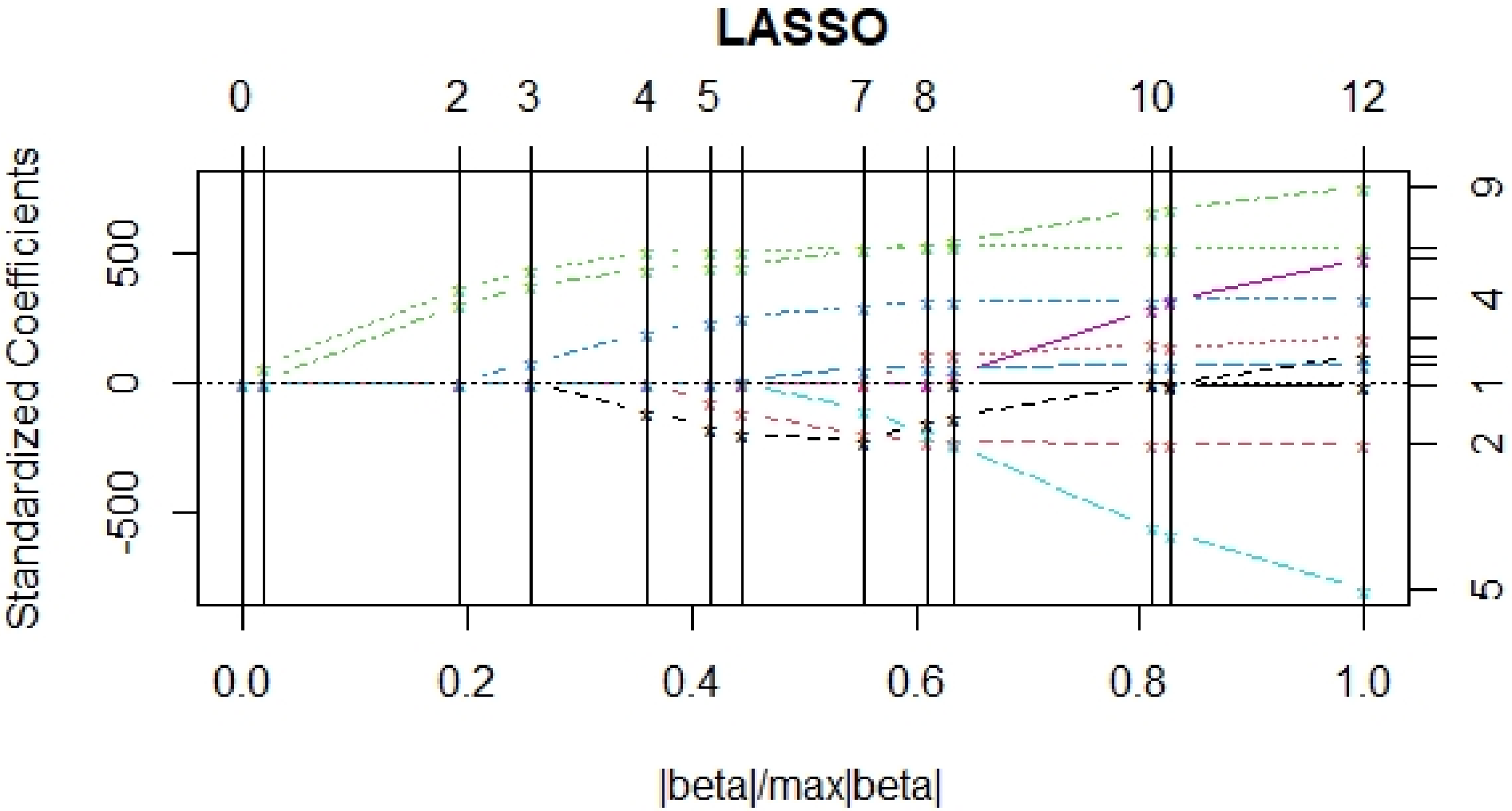
	AGE	SEX	BMI	BP	Serum measurements						Response
Patient	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	y
1	59	2	32.1	101	157	93.2	38	4	4.9	87	151
2	48	1	21.6	87	183	103.2	70	3	3.9	69	75
3	72	2	30.5	93	156	93.6	41	4	4.7	85	141
4	24	1	25.3	84	198	131.4	40	5	4.9	89	206
5	50	1	23.0	101	192	125.4	52	4	4.3	80	135
6	23	1	22.6	89	139	64.8	61	2	4.2	68	97
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
441	36	1	30.0	95	201	125.2	42	5	5.1	85	220
442	36	1	19.6	71	250	133.2	97	3	4.6	92	57

The response is a quantitative measure of disease progression one year after baseline.

[6] Efrom, B., Hastie, T., Johnstone, I. and Tibshirani, R. (2004) Least angel regression, Ann. Statist., 32(2), 407-499.

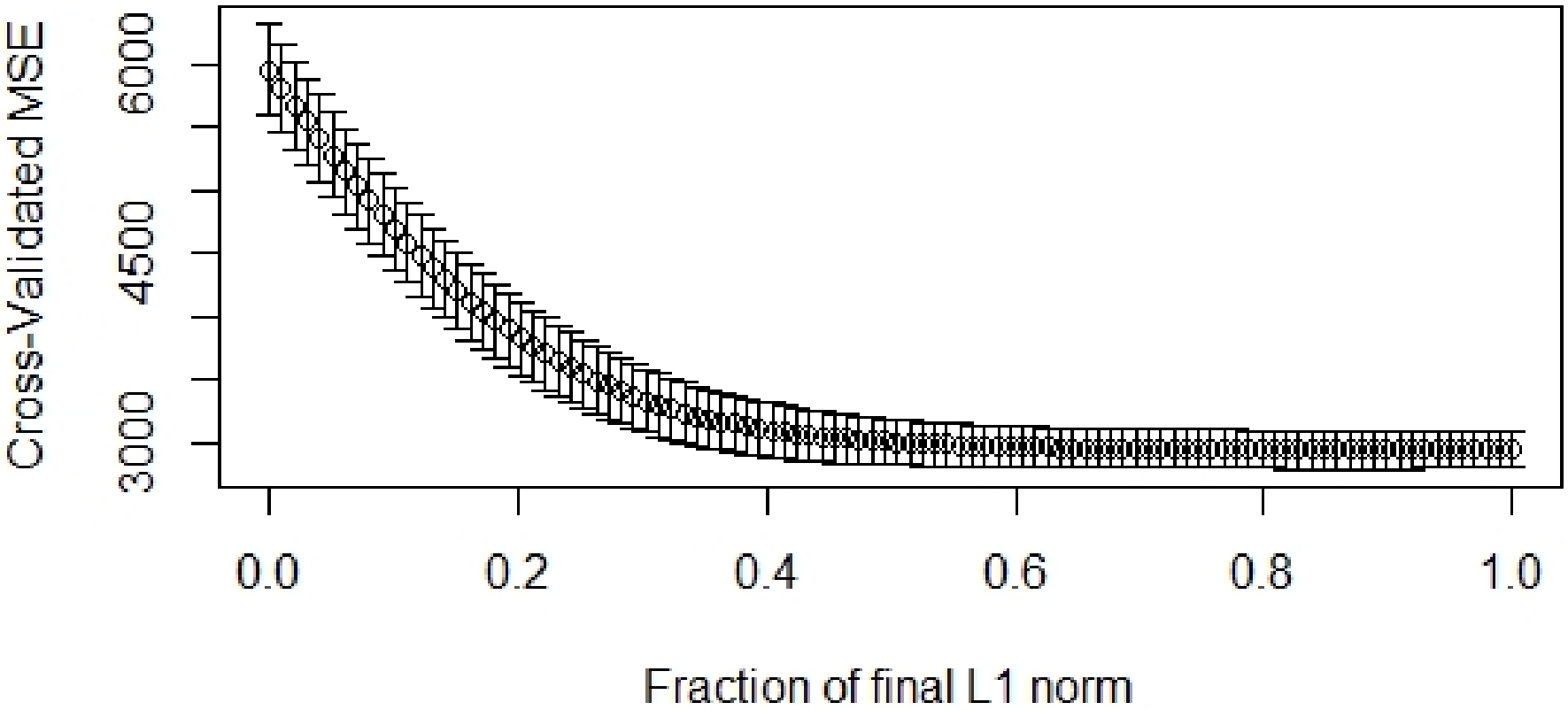
Lab13 cont.

Analyzing "Diabetes" data with "lars".



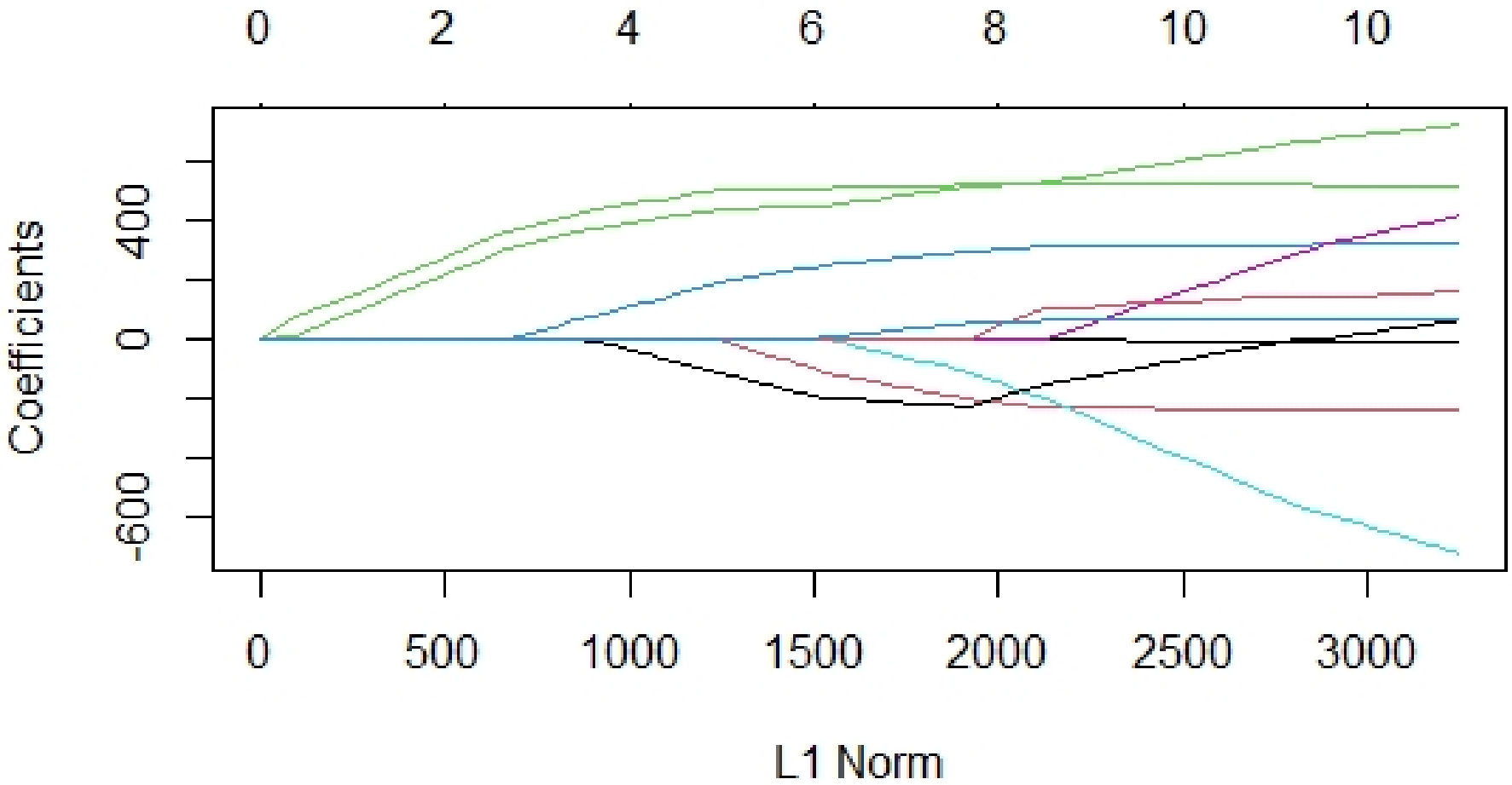
Lab13 cont.

Cross-validated mean squared prediction error:

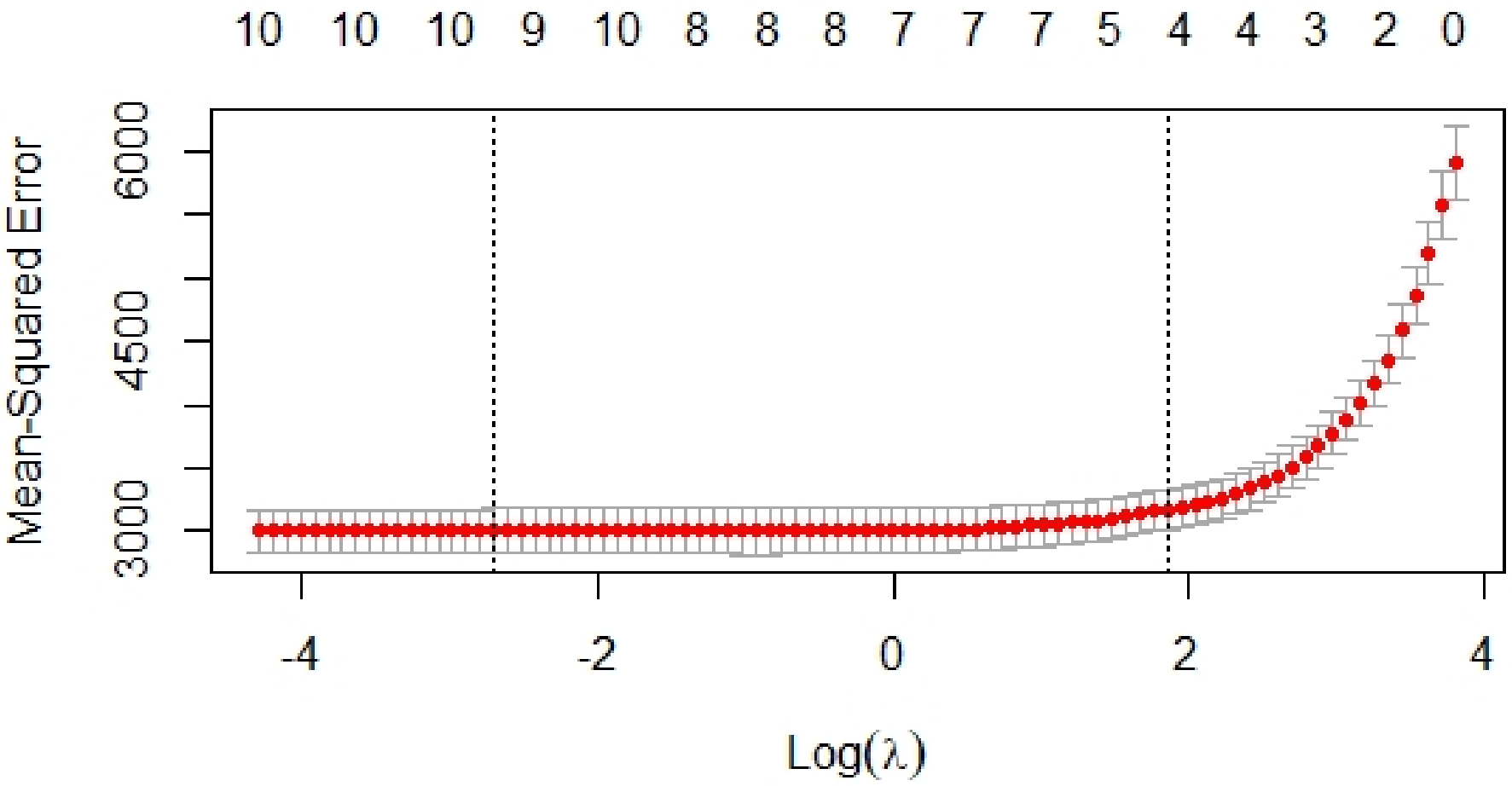


Lab13 cont.

Package "glmnet"



Lab13 cont.


10 10 10 9 10 8 8 8 7 7 7 5 4 4 3 2 0
Mean-Squared Error
3000
4500
6000
-4
-2
0
2
4
Log(λ)

Penalty functions

We want $p_\lambda(|\beta|) \geq p_\lambda(|\beta'|)$ when $|\beta| \geq |\beta'|$. This forces the optimization to set most coefficients in $\boldsymbol{\beta}$ to $\mathbf{0}$, thereby performing selection.

(Convex)Penalty	$p_\lambda(\lvert\beta_i\rvert)$	Description
Bridge	$\lambda\lvert\beta_i\rvert^\gamma$	$\gamma > 0$ $\gamma < 1$:unbiased $\gamma \leq 1$:sparse $\gamma > 1$:continues $\gamma \geq 1$:convex
Elastic net	$p_\lambda(\lvert\beta\rvert) = \lambda\left(\alpha\lvert\beta_i\rvert + (1-\alpha)\beta_i^2\right)$	$0 < \alpha < 1$
Adaptive LASSO	$\lambda\hat{w}_i\lvert\beta_i\rvert$, $\hat{w}_i = \frac{1}{\lvert\beta_i^*\rvert^\gamma}$	β_i^* is a $\sqrt{n}$-consistent estimator; $\gamma > 0$
Fused LASSO	$\lambda_1\lvert\beta_i\rvert + \lambda_2\lvert\beta_i - \beta_{i-1}\rvert$	

Penalty functions cont.

Concave penalties fulfill all three good properties.

Penalty	$p_\lambda(\lvert\beta_i\rvert)$	Description
SCAD	$\int_0^{\lvert\beta_i\rvert} \min\{1, \frac{(a-t/\lambda)_+}{a-1}\}dt = \begin{cases} \lambda\lvert\beta_i\rvert & \lvert\beta_i\rvert < \lambda \\ -\frac{\beta_i^2-2a\lambda\lvert\beta_i\rvert+\lambda^2}{2(a-1)} & \lambda < \lvert\beta_i\rvert \leq a\lambda \\ \frac{1}{2}(a+1)\lambda^2 & \lvert\beta_i\rvert > a\lambda \end{cases}$	$a > 2$
MCP	$\lambda \int_0^{\lvert\beta_i\rvert} \left(1 - \frac{t}{a\lambda}\right)_+ dt$	$a > 1$
Atan	$\lambda \left(\gamma + \frac{2}{\pi}\right) \arctan\left(\frac{\lvert\beta_i\rvert}{\gamma}\right)$	$\gamma > 0$
Mnet	$p_{\lambda_1}(\lvert\beta_i\rvert) + \lambda_2\beta_i^2$	

Lab14

Here, we fit the Enet (Elastic net) to a high-dimensional data.
Gene expression data (3051 genes and 38 tumor mRNA samples) from the leukemia microarray study

A list with the following elements:

`X`	a (38 x 3051) matrix giving the expression levels of 3051 genes for 38 leukemia patients. Each row corresponds to a patient, each column to a gene.
`Y`	a numeric vector of length 38 giving the cancer class of each patient.
`gene.names`	a matrix containing the names of the 3051 genes for the gene expression matrix `X`. The three columns correspond to the gene 'index', 'ID', and 'Name', respectively.

Lab15

Recall the analysis of ARMD data in Session 3, Lab7; specifically, the Assignment Q7: consider the following model

$$\begin{aligned}\text{VISUAL}_{it} &= \beta_0 + \beta_1 \text{VISUAL0}_i + \beta_2 \text{TIME}_{it} + \beta_3 \text{TREAT}_i \\ &\quad + \beta_4 \text{TREAT}_i \times \text{TIME}_{it} \\ &\quad + b_{0i} + b_{2i} \text{TIME}_{it} + \epsilon_{it}.\end{aligned}$$

We estimate the parameters using a "Penalized linear mixed-effects" procedure. Specifically, we use LASSO. SCAD can be also used.

Assignments

Q1 : Verify (1.1) and (1.2).

Q2 : Consider the model $y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \epsilon_i$ for $i = 1, \cdots, n$ and with the $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$. Suppose the parameter $\boldsymbol{\beta}$ is estimated by the ridge regression estimator as in (3.5).

a) The vector of "ridge residuals", defined as $\boldsymbol{\epsilon}(\lambda) = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{\mathrm{ridge}}(\lambda)$, are normally distributed. Why?

b) Show that $\mathrm{E}\boldsymbol{\epsilon}(\lambda) = \boldsymbol{A}\boldsymbol{X}\boldsymbol{\beta}$, where $\left[\boldsymbol{I}_n - \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X} + \lambda \boldsymbol{I}_p)^{-1}\boldsymbol{X}^\top\right]$.

c) Show that $\mathrm{Var}\,\boldsymbol{\epsilon}(\lambda) = \boldsymbol{A}^\top\boldsymbol{A}\sigma^2$.

d) Could the normal probability plot, i.e. a qq-plot with the quantiles of standard normal distribution plotted against those of the ridge residuals, be used to assess the normality of the latter? Motivate.

Assignments

Q3 : Consider the following penalized least squares criterion

$$PS_\lambda(\boldsymbol{\beta}) = \|y - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda \sum_{j=1}^{p} \hat{w}_i |\beta_j|, \quad \hat{w}_j = \frac{1}{|\hat{\beta}_j|},$$

where $\hat{\beta}_j$ is the jth element of the OLS estimator. For the case of orthonormal design, find the jth component of $\arg\min_{\boldsymbol{\beta}} PS_\lambda(\boldsymbol{\beta})$.

Thank you for your attention.